Python

Pandas

Day-1 :-

→ What is pandas?

It is an open source, where it is used in data analysis and data manipulation

$\boxed{\text{import pandas as pd}}$ → creating panda cor) gerating application

Example in real life:

Finding highest or average temperature in a week of a particular place in which the data is stored in a file.

→ What is data munging or data wrangling?

The process of cleaning data

* What is Dataframe in panda

It is object in pandas which used to represent with rows & columns

→ Syntax:

// For dictionary

If 'weather' is a dictionary then

```
df = pd.DataFrame(weather)
df
```

→ ~~Prior~~

calling a file

```
df = pd.read_csv("file name")
df
```

Checking dimension of a table

Syntax:- df.shape

If printing only rows then

step 1:-

rows, columns = df.shape

step 2:-

rows (Then no of rows will be printed)

columns (No of columns)

printing No of top rows

Syntax:-

df.head (#No of rows you want to print)

printing No of rows from bottom:-

Syntax:-

df.tail(#No of rows you want)

printing in between rows

df[start : end]

printing name of columns

df.columns

printing data of that columns

df.(Name of column you want)

printing maximum value of that column

Syntax :-

```
df['column_name'].max()
```

printing average or mean

Syntax :-

```
df['column_name'].mean()
```

printing minimum value of that column

Syntax :-

```
df['column_name'].min()
```

printing standard deviation

Syntax :-

```
df['column_name'].std()
```

printing statics of data

Syntax :-

```
df.describe()
```

output :-

column_name	
count	→ Total values
mean	→ mean
std	→ standard deviation
min	→ min
25%	→ 25% of data values below this value
50%	→ media (middle value)
75%	→ 75% of data value below this value
max	→ max value

printing rows that required to our requirem

Syntax

df[df.name then condition]

Example

df[df.temperature>=32]

Output:-

day	event	temp	windspeed
		32	6
		35	7
		32	4

printing only name instead of total row

Syntax:-

df[the value you want][df.temp==df[temp].max]

~~fo~~

printing two or more values you want

df[['day,temp']][df.temp==df[temp].max()]

must required

output:-

	day	temp
1	1/1/2017	45
2	2/1/2017	30

} This is a example data

Setting index instead of index 1,2,3 ...

Syntax:-

df.set_index('day', inplace=True)

↓

This is most because it change the original data

Process:-

Before

	day	event	temp	wind
0	1/1/2017	Rain	32	6
1	1/2/2017	Sunny	35	7

After

day	event	temp	wind
1/1/2017	Rain	32	6
1/2/2017	Sunny	35	7

df.loc['1/3/2017'] // printing the data you want

output

~~1/2/2017~~

Sc event : Rain

temp : 32

wind : 6

Reseting the index:

Syntax :-

df.reset_index(inplace=True)

setting index:

df.set_index('event', inplace=True)

Reading data in Files :-

if in csv file:

Syntax

```
df= pd.read_csv("file name")
df
```

if in excell file:

Syntax:

```
df= pd.read_excel("file name", "sheet no")
df
```